Module 7: Expanding Treatment & Design Structures

Nested Treatment Structure

Crossed vs Nested Treatment Structure

Crossed/Full Factorial

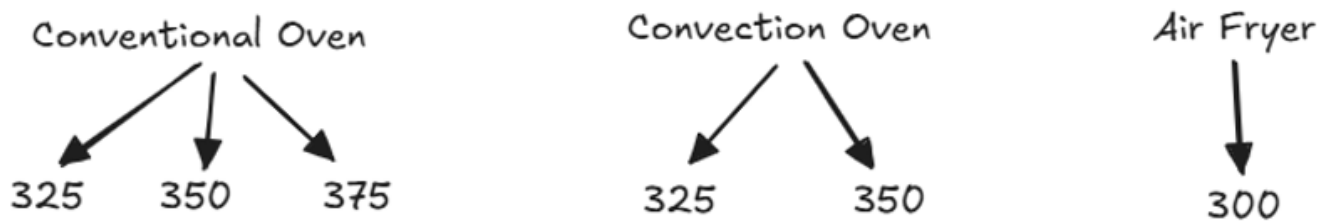
Each level of factor B appears with every level of Factor A

Nested

Levels of facotr B are unique within each level of Factor A

Example 7.4: Cookies in Air Fryers

> Baking is a precise science, and small changes in baking methods can have a significant impact on the final texture of cookies. A food scientist is conducting an experiment to determine how different baking methods influence cookie crispiness. However, not all baking methods allow for the same range of baking temperatures. For example, a conventional oven can accommodate a wider range of temperatures than an air fryer, which operates at lower heat settings. The food scientist bakes 6 cookies at each of the method and temperature combinations.



Therefore, we do not consider the main effect of Temperature. Instead, we will consider the individual effects of each Temperature within each Baking Method.

Statistical Effects Model for Nested Treatment Structures

$$y_{ijk} = \mu + \alpha_i + \beta(\alpha)_{ij} + \epsilon_{ijk} \text{ with } \epsilon_{ijk} \text{ iid} \sim N(0, \sigma^2)$$

for $i = 1, 2, 3; j = 1, 2, \ldots b_i; k = 1, 2, 3, 4, 5, 6$

where:

- y_{ijk}: the crispiness of the kth cookie baked at the jth temperature in the ith baking method.
- α_i: the effect of the ith baking method
- $\beta(\alpha)_{ij}$: the effect of the jth temperature nested within the ith baking method
- ϵ_{ijk}: the experimental error for the kth cookie baked at the jth temperature in the ith baking method.

ANOVA Table

SV	DF	SS	MS	F
A	(a-1)	SSA	MSA	MSA/MSE
B(A)	use subtraction	SSB(A)	MSB(A)	MSB(A)/MSE
e.u.(AB) - error	use subtraction	SSE	MSE	
Total	N-1			

Skeleton ANOVA

Source of Variation	DF

R: Analysis

```r
library(tidyverse)
cookie_data <- read_csv("data/07_cookie_data.csv") |>
  mutate(across(baking_method:cookie, as.factor))
head(cookie_data)
```

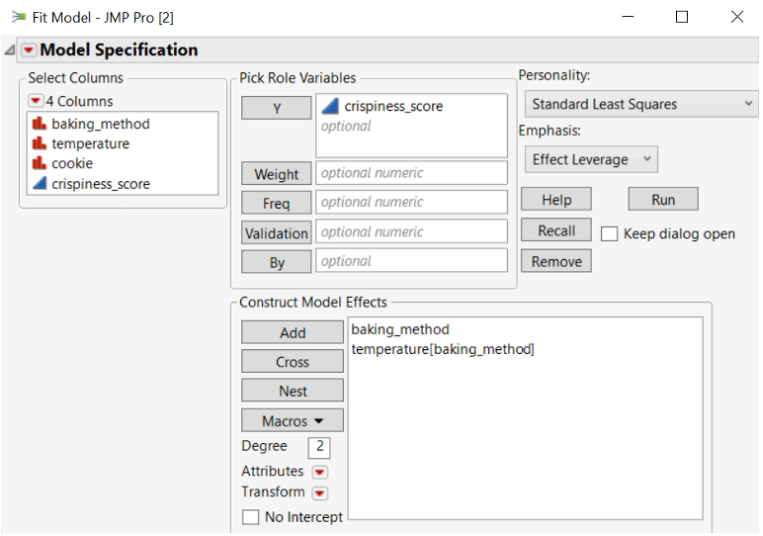
```
# A tibble: 6 × 4
  baking_method     temperature cookie crispiness_score
  <fct>             <fct>       <fct>             <dbl>
1 Conventional Oven 325         1                  69.0
2 Conventional Oven 325         2                  67.7
3 Conventional Oven 325         3                  69.3
4 Conventional Oven 325         4                  71.0
5 Conventional Oven 325         5                  67.5
6 Conventional Oven 325         6                  67.5
```

```r
cookie_mod <- lm(crispiness_score ~ baking_method + baking_method/temperature,
                 data = cookie_data)
anova(cookie_mod)
```

```
Analysis of Variance Table

Response: crispiness_score
                          Df  Sum Sq Mean Sq F value    Pr(>F)
baking_method              2 127.602  63.801 18.9852 4.699e-06 ***
baking_method:temperature  3  36.986  12.329  3.6687   0.02306 *
Residuals                 30 100.817   3.361
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

JMP: Analysis



Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	5	164.58810	32.9176	9.7953
Error	30	100.81677	3.3606	Prob > F
C. Total	35	265.40487		<.0001*

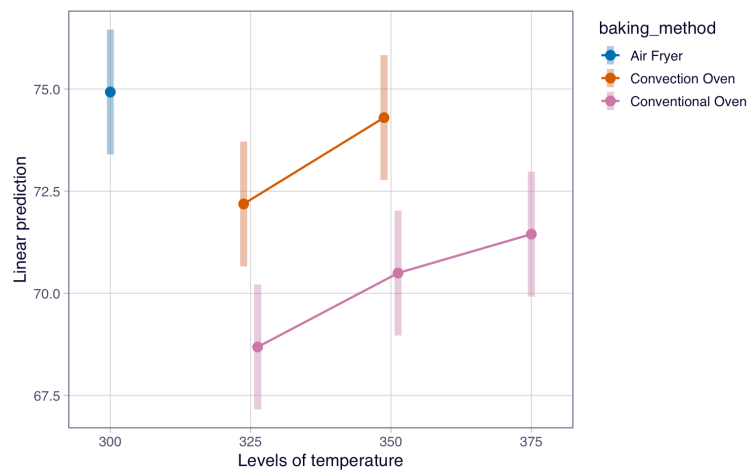
Parameter Estimates

Effect Tests

Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
baking_method	2	2	127.60195	18.9852	<.0001*
temperature[baking_method]	3	3	36.98615	3.6687	0.0231*

R: Results

```r
library(emmeans)
library(multcomp)
emmip(cookie_mod, baking_method ~ temperature,
      CIs = TRUE)
```

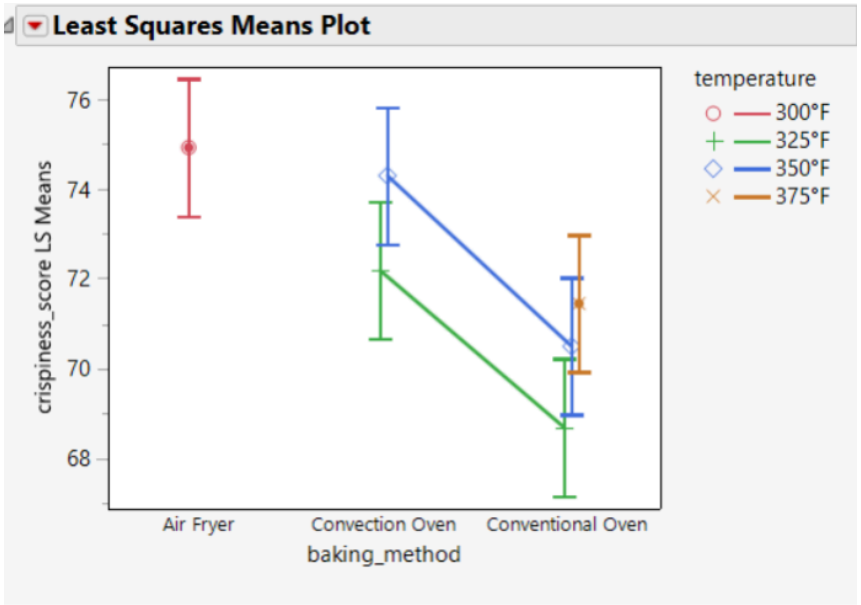



```r
emmeans(cookie_mod, specs = ~ baking_method:temperature) |>
  cld(Letters = LETTERS, decreasing = TRUE, adjust = "tukey")
```

```
 temperature baking_method     emmean    SE df lower.CL upper.CL .group
 300         Air Fryer           74.9 0.748 30     72.8     77.0  A
 350         Convection Oven     74.3 0.748 30     72.2     76.4  AB
 325         Convection Oven     72.2 0.748 30     70.1     74.3  ABC
 375         Conventional Oven   71.4 0.748 30     69.3     73.6   BCD
 350         Conventional Oven   70.5 0.748 30     68.4     72.6    CD
 325         Conventional Oven   68.7 0.748 30     66.6     70.8     D

Confidence level used: 0.95
Conf-level adjustment: sidak method for 6 estimates
P value adjustment: tukey method for comparing a family of 6 estimates
significance level used: alpha = 0.05
NOTE: If two or more means share the same grouping symbol,
      then we cannot show them to be different.
      But we also did not show them to be the same.
```

JMP: Results



Level					Least Sq Mean	Std Error
[Air Fryer]300°F	A				74.927	0.74839
[Convection Oven]350°F	A	B			74.300	0.74839
[Convection Oven]325°F	A	B	C		72.187	0.74839
[Conventional Oven]375°F		B	C	D	71.448	0.74839
[Conventional Oven]350°F			C	D	70.497	0.74839
[Conventional Oven]325°F				D	68.687	0.74839